

The missing boundary in stabilization of Grothendieck d -norms

Candidate correction to Conjecture 2.18 of arXiv:2003.07345

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Result. The rank bound preceding Conjecture 2.18 is one unit too large. The exact universal stabilization ranks for both symmetric Grothendieck norms are

$$\rho_{\mathbb{R}}(n) = \left\lfloor \frac{\sqrt{8n+1}-1}{2} \right\rfloor, \quad \rho_{\mathbb{C}}(n) = \lfloor \sqrt{n} \rfloor.$$

Consequently, every real parameter in $d(d+1)/2 \leq n < (d+1)(d+2)/2$ and every complex parameter in $d^2 \leq n < (d+1)^2$ is already an ordinary semidefinite program, although the conjecture places it in its NP-hard range. Block symmetrization gives the same unconditional polynomial-time overlap for the rectangular norm. Thus the conjectured frontier, under its natural uniform-parameter reading, cannot be correct unless $P = NP$. The strict range below the corrected threshold remains open here.

1 Source conjecture and the issue

Friedland and Lim introduce real and complex Grothendieck d -norms and prove a rank-stabilization result before asking whether the complementary parameter range is NP-hard [1]. Their Conjecture 2.18, on page 15, asserts NP-hardness for

$$\begin{array}{ll} \|\cdot\|_{\gamma,d}^{\mathbb{R}}, \|\cdot\|_{\Gamma,d}^{\mathbb{R}} & \text{when } d(d+1)/2 \leq n, \\ \|\cdot\|_{\mathbb{G},d}^{\mathbb{R}} & \text{when } d(d+1)/2 \leq m+n, \\ \|\cdot\|_{\gamma,d}^{\mathbb{C}}, \|\cdot\|_{\Gamma,d}^{\mathbb{C}} & \text{when } d^2 \leq n, \\ \|\cdot\|_{\mathbb{G},d}^{\mathbb{C}} & \text{when } d^2 \leq m+n. \end{array}$$

The preceding Lemma 2.16 uses

$$\left\lfloor \frac{\sqrt{8n+1}-1}{2} \right\rfloor + 1 \quad \text{and} \quad \lfloor \sqrt{n} \rfloor + 1$$

as upper bounds for extreme ranks. The final +1 is unnecessary. The proof below supplies the exact bounds and records the resulting correction to the complexity frontier.

2 Definitions

Let $\mathbb{K}$ be $\mathbb{R}$ or $\mathbb{C}$, and let $\text{Herm}_r(\mathbb{K})$ be the real vector space of real symmetric or complex Hermitian $r \times r$ matrices. For $n \geq 1$ define the elliptope and subelliptope

$$\mathcal{E}_n^{\mathbb{K}} = \{G \succeq 0 : g_{ii} = 1\}, \quad \mathcal{C}_n^{\mathbb{K}} = \{G \succeq 0 : 0 \leq g_{ii} \leq 1\}.$$

where the inequality follows from Lemma 2.16. The same argument applies to the (Γ, d) -norm. The equalities for the (G, d) -norm follows from (29). $\square$

The Grothendieck d -norms $\|\cdot\|_{\gamma,d}$, $\|\cdot\|_{\Gamma,d}$, $\|\cdot\|_{G,d}$ are NP-hard over $\mathbb{R}$ when $d = 1$; see (86) for example. We suspect the following:

Conjecture 2.18. The following norms are all NP-hard:

- (i) $\|\cdot\|_{\gamma,d}^{\mathbb{R}}$ and $\|\cdot\|_{\Gamma,d}^{\mathbb{R}}$ on $\mathbb{S}^n(\mathbb{R})$ when $d(d+1)/2 \leq n$;
- (ii) $\|\cdot\|_{G,d}^{\mathbb{R}}$ on $\mathbb{R}^{m \times n}$ when $d(d+1)/2 \leq m+n$;
- (iii) $\|\cdot\|_{\gamma,d}^{\mathbb{C}}$ and $\|\cdot\|_{\Gamma,d}^{\mathbb{C}}$ on $\mathbb{S}^n(\mathbb{C})$ when $d^2 \leq n$;
- (iv) $\|\cdot\|_{G,d}^{\mathbb{C}}$ on $\mathbb{C}^{m \times n}$ when $d^2 \leq m+n$.

It is perhaps worthwhile to point out that $\mathbb{I}_d^n$ and $\mathbb{G}_d^n$ are nonconvex sets when $d < n$, thus computing $\|\cdot\|_{\gamma,d}$ and $\|\cdot\|_{\Gamma,d}$ involve the nonconvex optimization problems in (26). On the other hand, when $d \geq n$, $\mathbb{I}_d^n = \mathbb{I}^n$ and $\mathbb{G}_d^n = \mathbb{G}^n$ are convex sets and the optimization problems in (26) are just semidefinite programs. An implication of Proposition 2.17 is that when d is in the range $\lfloor (-1 + \sqrt{1+8n})/2 \rfloor + 1 \leq d < n$ (for $\mathbb{R}$) or $\lfloor \sqrt{n} \rfloor + 1 \leq d < n$ (for $\mathbb{C}$), the nonconvex optimization problems in (26) are solvable in polynomial-time. For the case of $\|\cdot\|_{\gamma,d}^{\mathbb{R}}$, $d(d+1)/2 > n$, this follows from [32, Theorem 2.1], a result that originally appeared in [5, 38].

Figure 1: Proposition 2.17, Conjecture 2.18, and the stated polynomial-time range in the source paper, page 15.

The subscript d denotes the subset of matrices of rank at most d . For $A = A^*$, the source definitions are equivalently

$$\begin{aligned} \|A\|_{\gamma,d}^{\mathbb{K}} &= \max_{G \in \mathcal{E}_{n,d}^{\mathbb{K}}} |\operatorname{tr}(AG)|, \\ \|A\|_{\Gamma,d}^{\mathbb{K}} &= \max_{G \in \mathcal{C}_{n,d}^{\mathbb{K}}} |\operatorname{tr}(AG)|. \end{aligned}$$

Removing d means that the rank constraint is absent. For a rectangular $B \in \mathbb{K}^{m \times n}$ the source proves

$$\|B\|_{G,d}^{\mathbb{K}} = \frac{1}{2} \left\| \begin{bmatrix} 0 & B \\ B^* & 0 \end{bmatrix} \right\|_{\gamma,d}^{\mathbb{K}} = \frac{1}{2} \left\| \begin{bmatrix} 0 & B \\ B^* & 0 \end{bmatrix} \right\|_{\Gamma,d}^{\mathbb{K}}. \quad (1)$$

3 Proof intuition

Factor a rank- r correlation matrix as a Gram matrix of spanning vectors $v_1, \dots, v_n \in \mathbb{K}^r$. A small Hermitian perturbation H of the inner product preserves the diagonal precisely when $v_i^* H v_i = 0$ for every i . An extreme point cannot admit a nonzero such perturbation, so the rank-one matrices $v_i v_i^*$ must span the whole space of symmetric or Hermitian matrices. That space has real dimension $r(r+1)/2$ over $\mathbb{R}$ and r^2 over $\mathbb{C}$. Hence an extreme rank r obeys exactly the two displayed dimension inequalities.

Conversely, elementary vector configurations with those cardinalities have rank-one projectors spanning the full matrix space, so the bounds are sharp. Linear optimization over a compact convex set only needs its extreme points; therefore every rank constraint at least as large as the maximum extreme rank is redundant. This is the one-step correction missed by the source.

4 Exact symmetric stabilization

Set

$$\rho_{\mathbb{R}}(n) = \max\{r : r(r+1)/2 \leq n\}, \quad \rho_{\mathbb{C}}(n) = \max\{r : r^2 \leq n\}.$$

Lemma 1 (Extreme-rank bound). *Every extreme point of $\mathcal{E}_n^{\mathbb{K}}$ or $\mathcal{C}_n^{\mathbb{K}}$ has rank at most $\rho_{\mathbb{K}}(n)$.*

Proof. Let $G = VV^*$ have rank r , where the rows of V are spanning vectors $v_1, \dots, v_n \in \mathbb{K}^r$. Suppose first that G is extreme in $\mathcal{E}_n^{\mathbb{K}}$. If the real span of $v_1 v_1^*, \dots, v_n v_n^*$ is not all of $\text{Herm}_r(\mathbb{K})$, choose $0 \neq H \in \text{Herm}_r(\mathbb{K})$ orthogonal to every $v_i v_i^*$ in the trace inner product. For all sufficiently small $\varepsilon > 0$, $I \pm \varepsilon H \succeq 0$, and

$$G_{\pm} = V(I \pm \varepsilon H)V^*$$

are distinct positive semidefinite matrices with unit diagonal whose average is G . This contradicts extremality. The projectors therefore span, so

$$\dim_{\mathbb{R}} \text{Herm}_r(\mathbb{R}) = \frac{r(r+1)}{2} \leq n, \quad \dim_{\mathbb{R}} \text{Herm}_r(\mathbb{C}) = r^2 \leq n.$$

For G extreme in $\mathcal{C}_n^{\mathbb{K}}$, apply the same argument using only the active indices i for which $g_{ii} = 1$. If their projectors do not span, the active diagonal entries stay fixed under the perturbation above, while all inactive inequalities remain strict for sufficiently small ε . Again G splits nontrivially. There are at most n active indices, which gives the same bounds. $\square$

Lemma 2 (Sharpness). *For every r satisfying $r(r+1)/2 \leq n$ in the real case, or $r^2 \leq n$ in the complex case, $\mathcal{E}_n^{\mathbb{K}}$ has an extreme point of rank r . The same matrix is extreme in $\mathcal{C}_n^{\mathbb{K}}$.*

Proof. Over $\mathbb{R}$, take the $r(r+1)/2$ unit vectors

$$e_i \quad (1 \leq i \leq r), \quad \frac{e_i + e_j}{\sqrt{2}} \quad (1 \leq i < j \leq r).$$

Their rank-one projectors span every real symmetric matrix. Over $\mathbb{C}$, add the vectors $(e_i + ie_j)/\sqrt{2}$ for $i < j$; the resulting r^2 projectors span $\text{Herm}_r(\mathbb{C})$ over $\mathbb{R}$. If necessary, append copies of e_1 to obtain n vectors. To verify the converse perturbation criterion, suppose the Gram matrix G is the midpoint of G_+ and G_- in the elliptope. Positivity implies $\ker G \subseteq \ker G_+ \cap \ker G_-$, so $G_{\pm} = V(I \pm H)V^*$ for a Hermitian H . Unit diagonals give $v_i^* H v_i = 0$ for every i , and projector spanning forces $H = 0$. Thus the Gram matrix is extreme and has rank r .

If this matrix were a nontrivial convex combination of two subcorrelation matrices, every diagonal coordinate—an average of numbers at most one—could equal one only if both summands had unit diagonal. Extremality in the elliptope then makes the decomposition trivial. Thus it is also extreme in the subelliptope. $\square$

Theorem 1 (Exact universal threshold). *For $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$ and $d, n \geq 1$, the following are equivalent:*

1. $d \geq \rho_{\mathbb{K}}(n)$;
2. $\|A\|_{\gamma, d}^{\mathbb{K}} = \|A\|_{\gamma}^{\mathbb{K}}$ for every $A \in \text{Herm}_n(\mathbb{K})$;
3. $\|A\|_{\Gamma, d}^{\mathbb{K}} = \|A\|_{\Gamma}^{\mathbb{K}}$ for every $A \in \text{Herm}_n(\mathbb{K})$.

If $d < \rho_{\mathbb{K}}(n)$, there is an A with a strict gap in (2), and there is (possibly another) A with a strict gap in (3).

Proof. The extreme-rank lemma and finite-dimensional Krein–Milman theorem imply

$$\mathcal{E}_n^{\mathbb{K}} = \text{conv}(\mathcal{E}_{n,d}^{\mathbb{K}}), \quad \mathcal{C}_n^{\mathbb{K}} = \text{conv}(\mathcal{C}_{n,d}^{\mathbb{K}})$$

whenever $d \geq \rho_{\mathbb{K}}(n)$. Maximizing either sign of a linear functional gives (2) and (3).

Conversely, suppose $d < \rho_{\mathbb{K}}(n)$ and choose, by sharpness, an extreme correlation matrix G of rank larger than d . It is not in $\text{conv}(\mathcal{E}_{n,d}^{\mathbb{K}})$. In fact it is not in $\text{conv}(\mathcal{E}_{n,d}^{\mathbb{K}} \cup -\mathcal{E}_{n,d}^{\mathbb{K}})$: taking diagonals in such a representation would force every negatively signed term to have zero weight, after which extremality gives a contradiction. Strict separation of compact convex sets therefore produces a Hermitian A for which the full absolute support function is strictly larger than its rank- d counterpart. This is the required γ gap.

The same argument applies to $\mathcal{C}_{n,d}^{\mathbb{K}}$. Indeed, a convex combination of signed subcorrelation matrices can have every diagonal entry equal to one only when every used term is positively signed and itself has unit diagonal. Hence G is also outside the symmetrized rank- d subelliptope, and strict separation gives the Γ gap. $\square$

The extreme correlation rank characterization is classical [2, 3]; the proof is included to make both the strict threshold and the subcorrelation extension self-contained.

5 Consequences for Conjecture 2.18

The theorem is equivalent to the following corrected stabilization conditions:

$$\|\cdot\|_{\gamma,d}^{\mathbb{R}} = \|\cdot\|_{\gamma}^{\mathbb{R}}, \quad \|\cdot\|_{\Gamma,d}^{\mathbb{R}} = \|\cdot\|_{\Gamma}^{\mathbb{R}} \iff n < \frac{(d+1)(d+2)}{2}, \quad (2)$$

$$\|\cdot\|_{\gamma,d}^{\mathbb{C}} = \|\cdot\|_{\gamma}^{\mathbb{C}}, \quad \|\cdot\|_{\Gamma,d}^{\mathbb{C}} = \|\cdot\|_{\Gamma}^{\mathbb{C}} \iff n < (d+1)^2. \quad (3)$$

Here each equality means equality for every input matrix. The unrestricted quantities are support functions of spectrahedra and can be computed to arbitrary precision in polynomial time by two semidefinite programs, one for each sign of the objective.

For $B \in \mathbb{K}^{m \times n}$ let $N = m + n$. Combining (1) with the sufficient direction of the theorem gives

$$d \geq \rho_{\mathbb{K}}(N) \implies \|B\|_{\Gamma,d}^{\mathbb{K}} = \|B\|_{\Gamma}^{\mathbb{K}} \text{ for every } B. \quad (4)$$

This block argument is sufficient for the correction; no claim of an exact rectangular threshold is made.

Corollary 1 (Polynomial-time overlap). *All three real Grothendieck d -norms are semidefinite-programming quantities throughout the uniform parameter band*

$$\frac{d(d+1)}{2} \leq N < \frac{(d+1)(d+2)}{2},$$

where $N = n$ for the symmetric norms and $N = m + n$ for the rectangular norm. All three complex norms are likewise semidefinite-programming quantities throughout

$$d^2 \leq N < (d+1)^2.$$

These are infinite families lying inside all four clauses of Conjecture 2.18. Thus, under the standard assumption $P \neq NP$ and the natural uniform reading in which d may vary with the matrix dimension, the conjecture is false as stated.

For example, the symmetric norms already stabilize at $(d, n) = (2, 3)$ over $\mathbb{R}$ and at $(d, n) = (2, 4)$ over $\mathbb{C}$, exactly on the conjecture's claimed hard boundary. The correction is larger than these equality cases: it fills the entire bands above.

6 Complexity and scope caveats

There are three necessary qualifications.

1. A problem in polynomial time can also be NP-hard if $P = NP$. The unconditional statement proved here is SDP computability and the exact stabilization threshold; the word “counterexample” uses the conventional $P \neq NP$ interpretation also implicit in contrasting easy and hard ranges.
2. If “for fixed d ” was intended, and only $n \rightarrow \infty$ supplies an input family, the overlap contains only finitely many n for that particular d . Then this is a precise correction of the printed parameter frontier, not a refutation of possible fixed- d hardness in the remaining strict range. The paper’s own discussion allows d to scale with n , which is why the uniform reading is natural.
3. The proof gives arbitrary-precision polynomial-time SDP computation, matching the source paper’s formulation. It does not claim a rational closed form or exact bit-model output for generally algebraic optima.

The remaining possible hardness region is therefore

$$n \geq \frac{(d+1)(d+2)}{2} \quad (\mathbb{R}), \quad n \geq (d+1)^2 \quad (\mathbb{C})$$

for the symmetric norms, with analogous corrected sufficient boundaries in $m+n$ for the rectangular norms. No unconditional hardness result in that strict range is proved here.

7 Verification, novelty, and review recommendation

The proof was checked independently in four places: the perturbation preserves positive semidefiniteness and every active diagonal; the real dimensions are $r(r+1)/2$ and r^2 ; the explicit projector families span the required spaces; and diagonal coordinates exclude negative terms in the sharpness separation. Substitution at triangular and square boundary values confirms the one-step difference from Proposition 2.17.

A bounded novelty search covered the local run indexes, the exact paper title, “Conjecture 2.18”, “Grothendieck d -norms NP-hard”, “stabilization rank correlation matrices”, and later copies of the source paper. It found the classical extreme-correlation rank results and work on extreme bipartite correlations [4], but no published erratum or explicit correction of this conjectured frontier. Since the core rank theorem is classical and the correction is an arithmetic tightening of the source argument, novelty confidence is moderate, not high.

Human review should focus first on the intended complexity quantifiers in Conjecture 2.18, then on the exact-threshold converse using symmetrized convex hulls. The mathematical stabilization theorem itself is recommended as likely valid.

References

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